DNS (DOMAIN NAME SYSTEM) SERVER

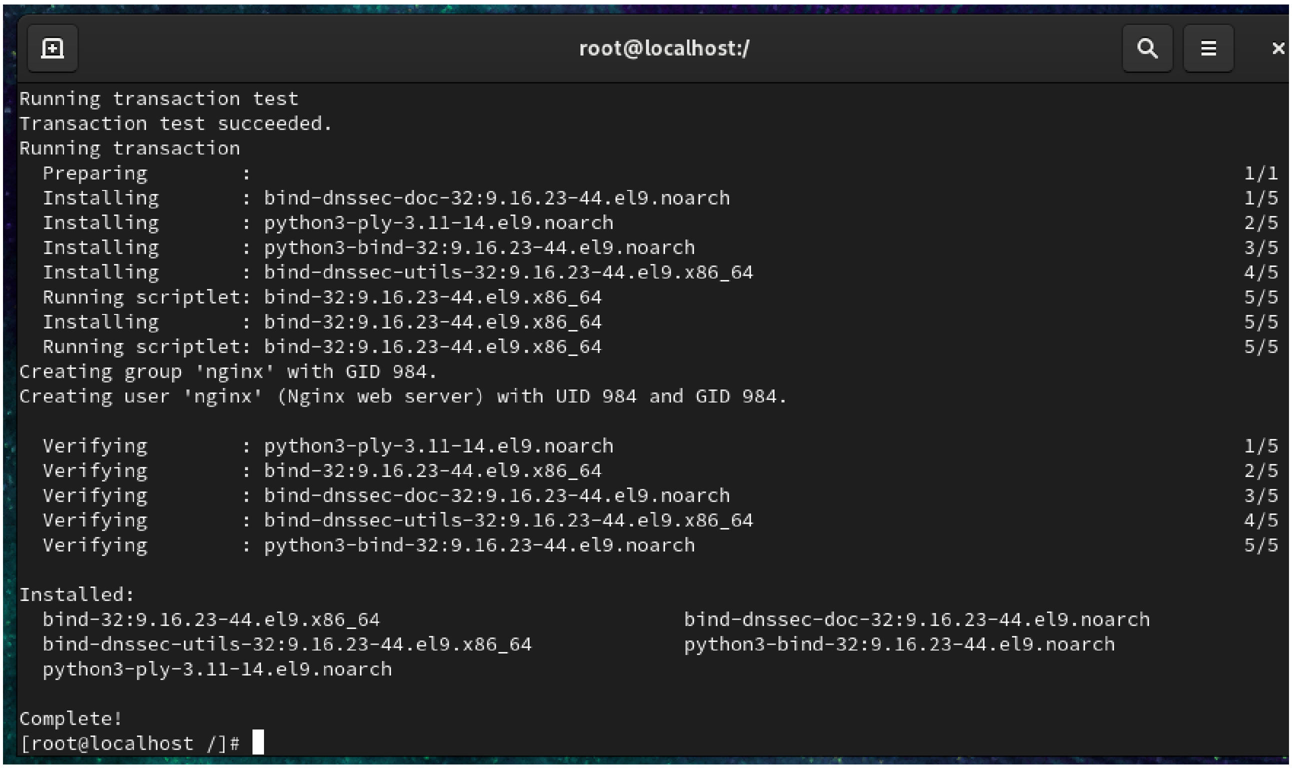
TASKS

1.Install BIND9 DNS server on your Linux machine and verify the installation by checking the version with the named -v command.

Install BIND9 DNS Server

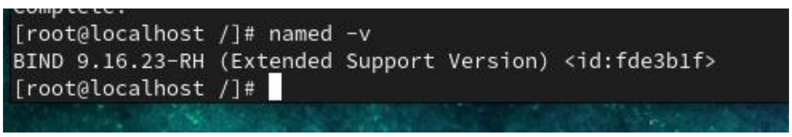
Step 1: Install BIND

dnf install bind bind-utils -y



Step 2: Check Version

named -v

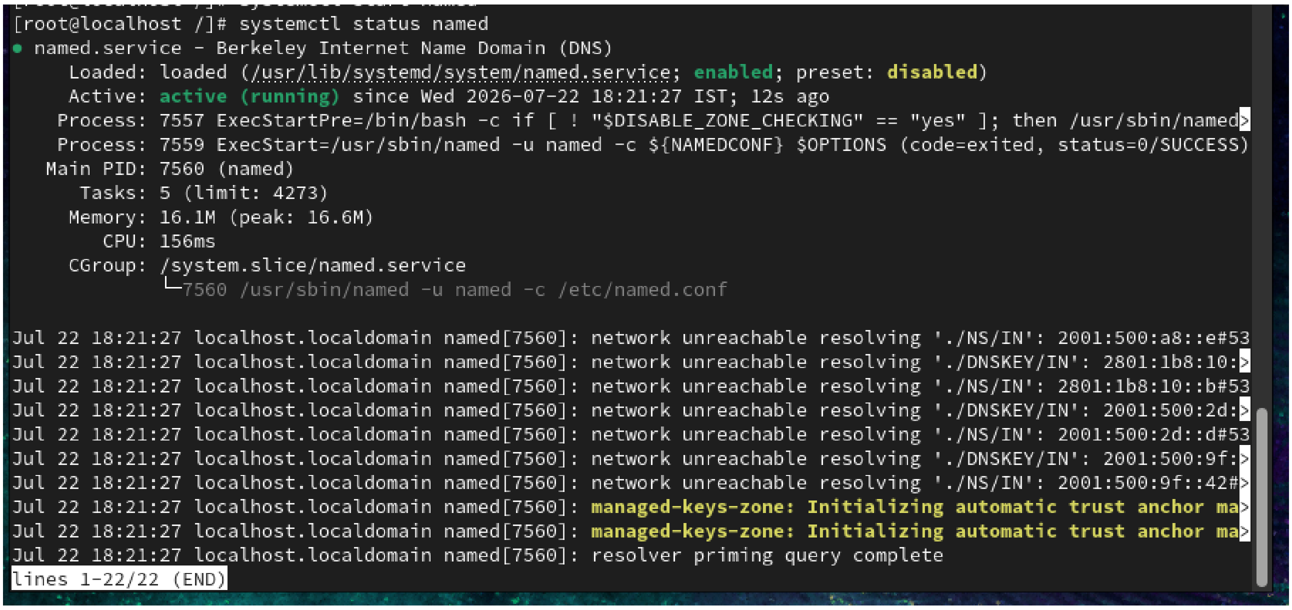


Step 3: Enable and Start DNS Service

systemctl enable named
systemctl start named

Check status:

systemctl status named



```
[root@localhost /]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-07-22 18:21:27 IST; 12s ago
    Process: 7557 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/sbin/named>
    Process: 7559 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 7560 (named)
      Tasks: 5 (limit: 4273)
     Memory: 16.1M (peak: 16.6M)
        CPU: 156ms
     CGroup: /system.slice/named.service
             └─7560 /usr/sbin/named -u named -c /etc/named.conf

Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './NS/IN': 2001:500:a8::e#53
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10:>
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './DNSKEY/IN': 2001:500:2d:>
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './NS/IN': 2001:500:2d::d#53
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './DNSKEY/IN': 2001:500:9f:>
Jul 22 18:21:27 localhost.localdomain named[7560]: network unreachable resolving './NS/IN': 2001:500:9f::42#>
Jul 22 18:21:27 localhost.localdomain named[7560]: managed-keys-zone: Initializing automatic trust anchor ma>
Jul 22 18:21:27 localhost.localdomain named[7560]: managed-keys-zone: Initializing automatic trust anchor ma>
Jul 22 18:21:27 localhost.localdomain named[7560]: resolver priming query complete
lines 1-22/22 (END)
```

2.Configure a basic DNS zone file in /etc/bind/ for a custom domain like myzomatoclone.local, mapping it to 127.0.0.1, and restart the DNS service to apply changes.

Hint: Edit named.conf.local and create a new zone file using db.local as a template.

Create DNS Zone

Step 1: Backup Configuration

cp /etc/named.conf /etc/named.conf.bak

Step 2: Edit named.conf

vi /etc/named.conf

Find:

listen-on port 53 { 127.0.0.1; };

Change to:

listen-on port 53 { any; };

Find:

allow-query { localhost; };

Change to:

allow-query { any; };

Step 3: Add Zone

Go to the bottom and add:

```
zone "myzomatoclone.local" IN {
    type master;
    file "db.myzomatoclone.local";
};
```

Save:

```
Esc
:wq
```

Step 4: Create Zone File

cp /var/named/named.localhost /var/named/db.myzomatoclone.local

---

Step 5: Edit Zone File

vi /var/named/db.myzomatoclone.local

Replace everything with:

```
$TTL 1D

@   IN SOA  ns1.myzomatoclone.local. root.myzomatoclone.local. (

       2026071401

       1H

       15M

       1W

       1D )


@       IN NS      ns1.myzomatoclone.local.

ns1     IN A       127.0.0.1
```

@ IN A 127.0.0.1

Save:

Esc

:wq

Step 6: Set Permissions

chown root:named /var/named/db.myzomatoclone.local

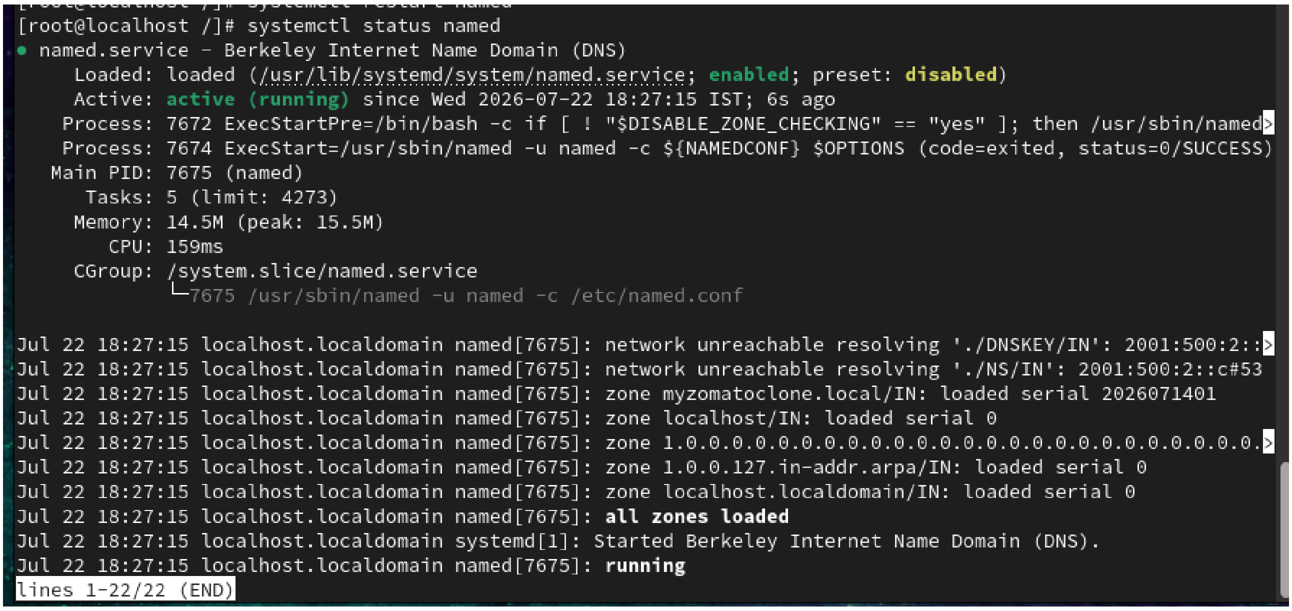
chmod 640 /var/named/db.myzomatoclone.local

Step 7: Restart DNS

systemctl restart named

Check:

systemctl status named



```
[root@localhost /]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-07-22 18:27:15 IST; 6s ago
    Process: 7672 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/sbin/named>
    Process: 7674 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 7675 (named)
      Tasks: 5 (limit: 4273)
     Memory: 14.5M (peak: 15.5M)
        CPU: 159ms
     CGroup: /system.slice/named.service
             └─7675 /usr/sbin/named -u named -c /etc/named.conf

Jul 22 18:27:15 localhost.localdomain named[7675]: network unreachable resolving './DNSKEY/IN': 2001:500:2::>
Jul 22 18:27:15 localhost.localdomain named[7675]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Jul 22 18:27:15 localhost.localdomain named[7675]: zone myzomatoclone.local/IN: loaded serial 2026071401
Jul 22 18:27:15 localhost.localdomain named[7675]: zone localhost/IN: loaded serial 0
Jul 22 18:27:15 localhost.localdomain named[7675]: zone 1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.>
Jul 22 18:27:15 localhost.localdomain named[7675]: zone 1.0.0.127.in-addr.arpa/IN: loaded serial 0
Jul 22 18:27:15 localhost.localdomain named[7675]: zone localhost.localdomain/IN: loaded serial 0
Jul 22 18:27:15 localhost.localdomain named[7675]: all zones loaded
Jul 22 18:27:15 localhost.localdomain systemd[1]: Started Berkeley Internet Name Domain (DNS).
Jul 22 18:27:15 localhost.localdomain named[7675]: running
lines 1-22/22 (END)
```

3.Set your system's DNS resolver to use your local BIND9 server and test domain resolution for myzomatoclone.local using the dig or nslookup command.

Configure Resolver

Step 1: Edit resolv.conf

vi /etc/resolv.conf

Add:

nameserver 127.0.0.1

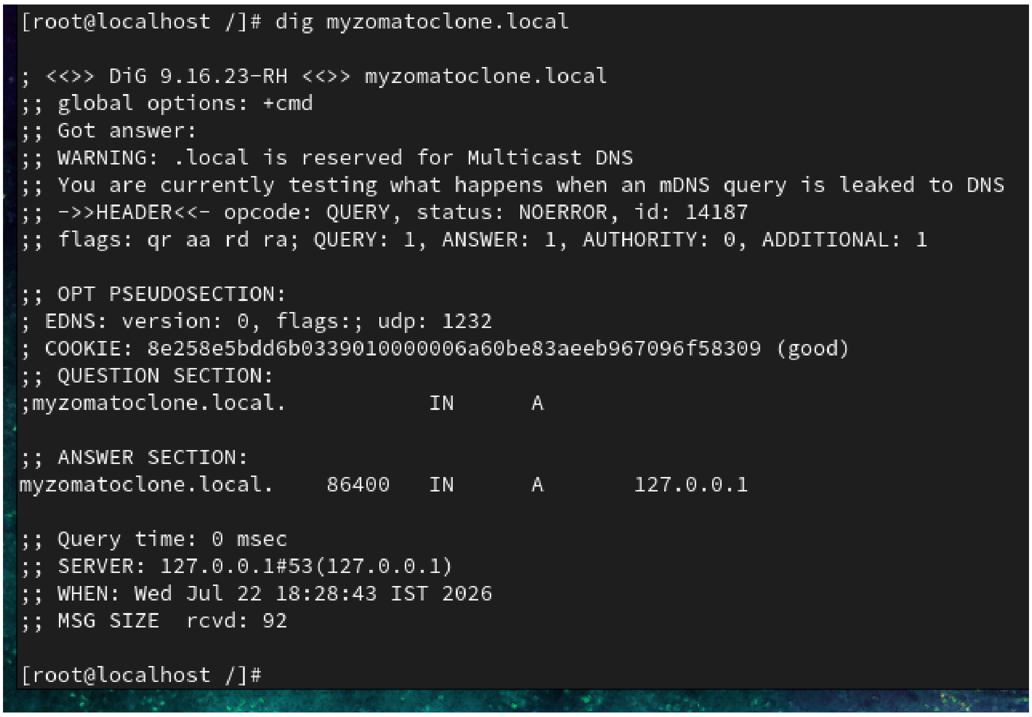
Save.

Step 2: Test DNS

nslookup myzomatoclone.local

Also test:

dig myzomatoclone.local



```
[root@localhost /]# dig myzomatoclone.local

; <<>> DiG 9.16.23-RH <<>> myzomatoclone.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 14187
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8e258e5bdd6b0339010000006a60be83aeeb967096f58309 (good)
;; QUESTION SECTION:
;myzomatoclone.local.           IN      A

;; ANSWER SECTION:
myzomatoclone.local.    86400   IN      A       127.0.0.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1)
;; WHEN: Wed Jul 22 18:28:43 IST 2026
;; MSG SIZE  rcvd: 92

[root@localhost /]#
```

4.Add a CNAME record in your zone file to create an alias foodsearch.myzomatoclone.local that points to myzomatoclone.local, then verify the alias works with dig.

Add CNAME Record

Edit zone:

vi /var/named/db.myzomatoclone.local

Add:

foodsearch IN CNAME myzomatoclone.local.

Save.

Increase serial number.

Change:

2026071401

to

2026071402

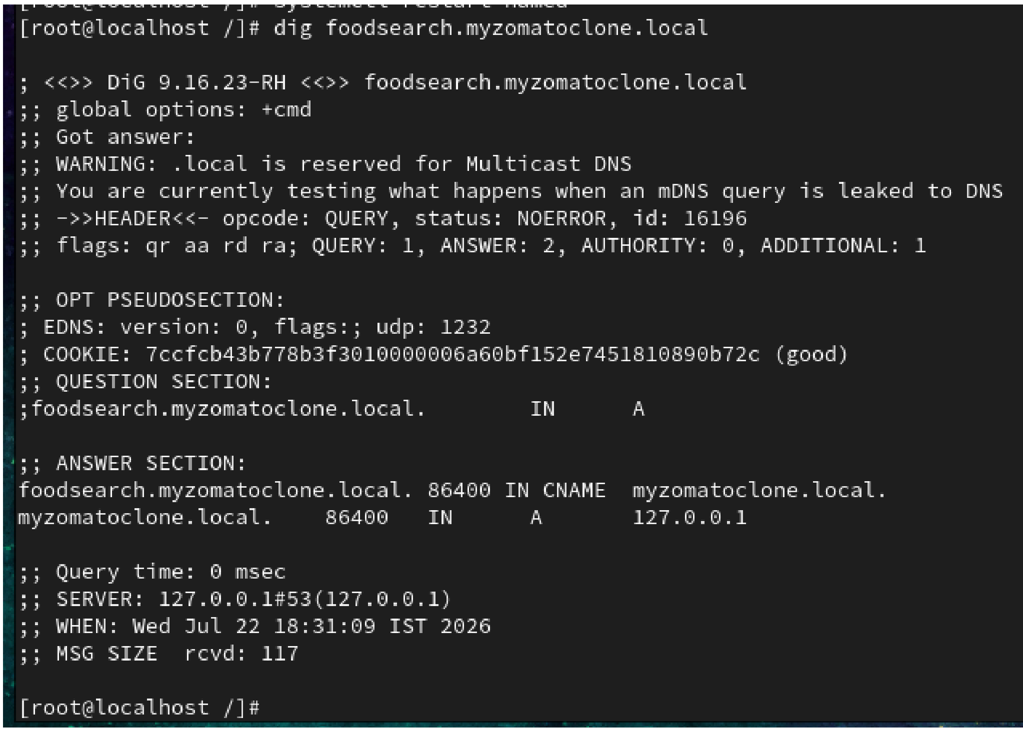
Save.

Restart DNS:

systemctl restart named

Verify:

dig foodsearch.myzomatoclone.local



5.Find and fix the error in this zone file snippet: 'myzomatoclone.local. IN A 127.0.0.1 foodsearch IN CNAME myzomatoclone.local.' — the CNAME is not resolving as

**expected.

Hint: Check the syntax and placement of the CNAME record.**

Fix Zone Error

Incorrect:

myzomatoclone.local. IN A 127.0.0.1
foodsearch IN CNAME myzomatoclone.local.

Correct:

@ IN A 127.0.0.1
foodsearch IN CNAME myzomatoclone.local.

Check syntax:

named-checkconf

Check zone:

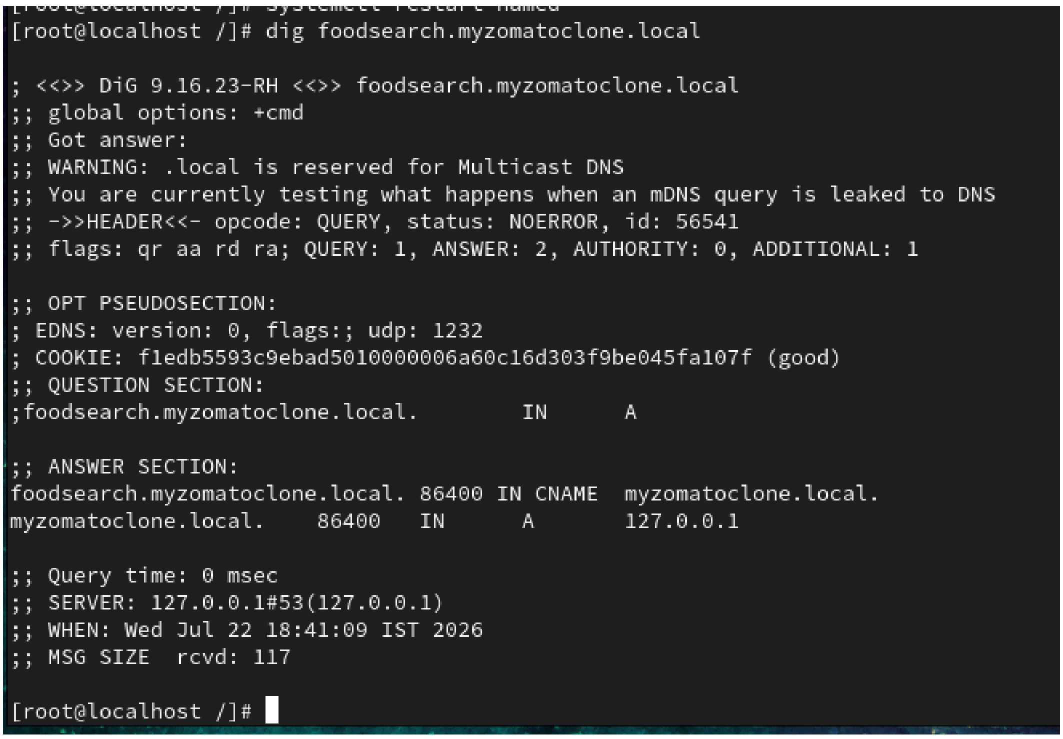
named-checkzone myzomatoclone.local /var/named/db.myzomatoclone.local

Restart:

systemctl restart named

Verify:

dig foodsearch.myzomatoclone.local



```
[root@localhost /]# dig foodsearch.myzomatoclone.local

; <<>> DiG 9.16.23-RH <<>> foodsearch.myzomatoclone.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56541
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: f1edb5593c9ebad5010000006a60c16d303f9be045fa107f (good)
;; QUESTION SECTION:
;foodsearch.myzomatoclone.local.        IN      A

;; ANSWER SECTION:
foodsearch.myzomatoclone.local. 86400 IN CNAME  myzomatoclone.local.
myzomatoclone.local.    86400   IN      A       127.0.0.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1)
;; WHEN: Wed Jul 22 18:41:09 IST 2026
;; MSG SIZE  rcvd: 117

[root@localhost /]#
```